Plausible, Scrutinized, Dismantled: A Rigorous Negative Result for Physiology-Constrained Cross-Hospital Transfer

Anonymous Author(s)

Abstract

Physiology-constrained pretraining does not improve cross-hospital sepsis prediction. A well-motivated hypothesis – that penalizing violations of known physiological relationships during masked-value pretraining of a clinical time-series transformer would yield representations that transfer better across hospitals than an unconstrained baseline – initially appeared confirmed: the constrained model beat empirical risk minimization (ERM) at every held-out site. Systematic re-evaluation traced the entire apparent gain to five identifiable evaluation confounds. Correcting all five reverses the result: the constrained model falls below ERM at every site, and a plain gradient-boosted-tree baseline on aggregate features matches or beats both neural methods. Run-to-run seed variance of 4 to 7 AUROC points exceeds most of the originally reported effect sizes. The negative result also holds under a second, independently designed test: a label-free model-selection criterion built on the same physiological signal fails to reliably outperform a simple unsupervised baseline. We report each confound mechanistically and in generalizable terms, so a reader building a different domain-knowledge-constrained clinical method can recognize the same traps before publishing them.

Keywords: physiology-constrained learning, negative results, cross-hospital generalization, evaluation confounds, clinical machine learning, reproducibility

Data and Code Availability All experiments use three publicly available, de-identified, credentialed-access ICU databases: the PhysioNet/Computing in Cardiology Challenge 2019 dataset (Reyna et al., 2020), MIMIC-IV (Johnson et al., 2023), and the eICU Collaborative Research Database (Pollard et al., 2018). Code – including the corrected SOFA-based label-derivation pipeline and the source-domain-only evaluation pipeline central to Section 4 – will be made available as an anonymized repository at the time of submission for review.

Institutional Review Board (IRB) This work is a secondary analysis of pre-existing, de-identified, credentialed-access datasets already approved for secondary research use under their respective data use agreements. No new human-subjects data was collected, and no additional IRB approval was required.

1. Introduction

Clinical machine learning models trained at one hospital routinely lose accuracy when deployed at another, and the consequences are not hypothetical. An external validation of a proprietary sepsis prediction model widely deployed across US hospitals found an AUROC of 0.63, a sensitivity of 33%, and a positive predictive value of 12% at a single academic medical center – well below the vendor's reported performance – contributing to the model's deactivation at that site (Wong et al., 2021).

This paper tests a specific, principled response to that problem: physiology-constrained learning (PCL), which augments masked-value pretraining of a clinical time-series transformer with differentiable penalties for violating known physiological relationships among simultaneously measured variables. The premise follows the Independent Causal Mechanisms principle (Schölkopf et al., 2021): relationships such as the mean arterial pressure identity or the Henderson-Hasselbalch equation hold at every hospital regardless of local measurement device, analyzer calibration, or ordering practice, so a representation constrained to respect them should depend less on site-specific artifacts than one that is not. This is not a strawman hypothesis; it is the response a careful practitioner reaches for when asked to build a clinical model that generalizes across sites, and it is the hypothesis an earlier version of this investigation set out to confirm.

Our contribution is not the claim that this hypothesis failed; it is a complete, rigorous diagnostic accounting of why a principled hypothesis produced a false positive convincing enough to submit, general enough that a reader building a different domain-knowledge-constrained method can recognize the same traps in their own pipeline before they reach publication. An earlier version of this investigation was submitted elsewhere and withdrawn after our own follow-up investigation uncovered the confounds this paper documents; we report that investigation in full here.

2. Method

PCL pretrains a six-layer clinical time-series transformer (d=256, 8 heads, ∼3.2M parameters) on 48-hour, hourly-binned ICU windows using a masked-value prediction objective, following the general masked-pretraining paradigm (Vaswani et al., 2017; Devlin et al., 2019). The training objective is $L = L_{\text{mask}} + \lambda L_{\text{phys}}$, where L_{phys} averages differentiable residuals on three physiological relations – an L_1 penalty on the mean arterial pressure identity and squared-error penalties on the Henderson-Hasselbalch and Severinghaus equations – applied only at timesteps where all required variables were observed and at least one was masked, so the penalty must be satisfied by imputation from cross-variable structure rather than by copying the input. Models are trained on PhysioNet Site A (10,381 stays, 3.3% sepsis prevalence) and evaluated zero-shot on PhysioNet Site B (14,779 stays, 2.1%), MIMIC-IV (74,607 stays, 30.9%), and eICU-CRD (130,446 stays, 8.8%) on the task of sepsis onset prediction. The corrected pipeline (Section 5) uses 17 physiological variables, up from the 9 used in the original result reported next; the expansion improved external transfer for both ERM and PCL and was retained throughout, so comparing Table 1 against Table 2 is comparing two different input representations, not the same model re-scored under corrected labels alone.

3. The Initial, Misleading Result

Table 1 reports the original, first-pass result: PCL appears to beat every baseline at every held-out site, with the margin growing with distributional distance – +6.4 percentage points at Site B, +10.1 at MIMIC-IV, +8.4 at eICU – and PCL is the best method in every one of three seeds at every site. The result also passed a constraint-randomization ablation (real physiological constraints outperformed shuffled, wrong-equation, and noise-injected variants, with the two most-corrupted variants falling below unconstrained ERM) and a measurement-noise robustness check (PCL's AUROC degraded less than ERM's under injected sensor bias). Nothing about the result looked fragile from any angle tried at the time; that is precisely what makes the diagnosis in Section 4 worth reporting in full.

Table 1: The original, uncorrected result: zero-shot cross-hospital sepsis AUROC, mean over 3 seeds. PCL appears to beat every baseline at every held-out site. This result does not survive Section 4 and is reported here only to show how convincing it looked.

Method	Source	Site B	MIMIC-IV	eICU
ERM	0.823	0.697	0.582	0.541
DRO	0.778	0.683	0.612	0.604
IRM†	—	0.675	—	—
PCL	**0.865**	**0.761**	**0.682**	**0.626**

[†IRM: single seed, Site B only, k=3 care-unit environments. Margins as originally reported: +6.4pp Site B, +10.1pp MIMIC-IV, +8.4pp eICU.]

4. The Five Confounds

Each confound below is stated mechanistically and in terms general enough that a reader building a different domain-knowledge-constrained method can check their own pipeline against it; each ends with the generalized lesson.

4.1. Label-Definition Shift

Training labels followed the clinical Sepsis-3 consensus definition (Singer et al., 2016), while external evaluation on MIMIC-IV and eICU originally used administrative ICD-code-derived sepsis labels. Clinical and claims-based sepsis definitions are known to diverge substantially (Rhee et al., 2017), so cross-site AUROC differences conflated genuine distribution shift with a change in the definition of the outcome itself. We corrected this by re-deriving Sepsis-3 directly via SOFA scoring anchored to a suspected-infection window, following the standard reference implementation (Seymour et al., 2016), yielding sepsis prevalence of 30.9% on MIMIC-IV and 8.8% on eICU. **Lesson:** any cross-site comparison must confirm the outcome definition, not only the input distribution, is held fixed across sites.

4.2. Pretraining Data Leakage Disguised as Zero-Shot Transfer

PCL's masked-value pretraining used unlabeled data from all four sites, while ERM's pretraining used source-site data only. Although pretraining is self-supervised and touches no labels, exposure to target-site inputs during pretraining still breaks the zero-shot premise the comparison depends on: PCL had, in effect, already seen the deployment distribution. **Lesson:** "no labels touched" is not the same guarantee as "no target data touched"; a zero-shot claim must audit every stage of training, not only the labeled fine-tuning stage.

4.3. OOD-Contaminated Hyperparameter Selection

The constraint weight λ, which strongly affects cross-hospital performance, was originally chosen using AUROC on a held-out target site rather than on source-domain validation data, so the reported zero-shot numbers were selected with knowledge of the test distribution they were meant to predict. **Lesson:** a hyperparameter is only as zero-shot as the data used to select it; auditing final-model performance is not sufficient if the selection procedure that produced that model touched target data.

4.4. Partial Circularity in the Severinghaus Constraint

PhysioNet does not record arterial oxygen partial pressure (PaO2) directly; the released dataset reconstructs it from oxygen saturation by inverting the Severinghaus equation – the same relation PCL's constraint penalizes the model for violating. The constraint's training-time supervision was therefore partly self-referential: it could be satisfied, in part, by recovering a value the data-generating process had already computed from the constraint's own equation. **Lesson:** before enforcing a domain-knowledge constraint, audit whether any of its input variables were themselves derived by assuming that constraint; a preprocessing pipeline can silently manufacture the invariance a model is later credited with discovering.

4.5. Measurement-Composition Artifact in the Aggregate Violation Score

The aggregate physiological violation score averages whichever of the three constraints happen to be computable at a given timestep, and constraint availability is highly uneven and site-dependent – for instance, bicarbonate is charted in 8.1% of Site A stay-hours versus 0.16% at Site B, a fifty-fold difference driven by local lab-ordering practice, not physiology. Because the sparser constraints also occupy a different residual scale than the dense ones, a site's aggregate mean partly tracks which labs are ordered there rather than any physiological difference. Recomputing each site's aggregate with pooled component values, on raw data with no model involved, isolates roughly 49% of the cross-site gap as composition artifact and the remaining 51% as a genuine per-component difference – partially, not wholly, artifactual. **Lesson:** an aggregate score over heterogeneously-available components is a composition statistic before it is anything else; decompose it before reading a cross-site difference in it as a physiological effect.

5. The Corrected Result

With all five confounds addressed, PCL falls below ERM at every site (Table 2), alongside a group-distributionally-robust baseline (Sagawa et al., 2020). The Source column also moves between the two tables for both methods, not only the target sites; this is expected. Two changes bundled into the correction affect in-domain fit directly: the 9-to-17-variable expansion (Section 2) and a different λ-selection procedure (below), so the corrected PCL model is a genuinely different trained model, not the same model re-scored.

A gradient-boosted tree (Chen and Guestrin, 2016) trained on simple per-stay summary statistics matches or exceeds both neural methods at every external site (0.754 / 0.654 / 0.660 versus ERM's 0.701 / 0.649 / 0.592 and PCL's 0.655 / 0.622 / 0.572), which further questions whether the transformer architecture is earning its complexity here. We separately checked whether invariant risk minimization (Arjovsky et al., 2019) with 30 real hospital environments drawn from eICU closes the gap to ERM; it does not, but that check turned out not to support any claim about this paper's zero-shot setting, for two reasons, and we report it only for completeness. First, that comparison is in-distribution within eICU, not zero-shot Site-A-to-eICU transfer, so it does not speak to the transfer failure this paper is about. Second, the gap it found (0.788 versus its own ERM baseline of 0.792, 0.4 percentage points) is smaller than this study's own run-to-run seed noise (4 to 7 percentage points, below), so it is not distinguishable from noise in the first place. We do not treat this as evidence either way.

Run-to-run variability across seeds reached 4 to 7 AUROC points on this pipeline, exceeding most of the differences originally reported as method effects in Table 1: a single-seed comparison at this scale is not informative regardless of how large the claimed gain looks.

The constraint weight λ itself was selected by sweeping six candidate values and taking the one with the highest AUROC on a held-out validation split of the source site (PhysioNet Site A) only – the same split used for every other hyperparameter decision, never a target site. This replaces the original selection procedure named in Section 4.

Table 2: Corrected cross-hospital sepsis AUROC, mean over 3 seeds, all five confounds addressed. PCL is now below ERM at every site.

Method	Source	Site B	MIMIC-IV	eICU
ERM	0.818	0.701	0.649	0.592
PCL	0.798	0.655	0.622	0.572
Group DRO	0.822	0.717	0.688	0.628
XGBoost	—	0.754	0.654	0.660

6. Beyond Prediction Accuracy

The same physiological signal was also tested as a label-free criterion for selecting the constraint weight without target labels – an independently designed test, since a selection criterion can fail even when the underlying representation does not, and vice versa. Prior work by the same author (Tamvada and Lin, 2026) evaluates this directly and finds that the physiology-violation criterion does not reliably outperform a simple unsupervised baseline (reconstruction error) at this task. That work is itself unpublished and under review, so we report only the qualitative outcome here rather than its numbers, and do not reproduce its tables; see it for the full comparison and the validation-failure diagnosis motivating it.

A calibration or conformal-prediction-coverage test was considered but is not included here: no such artifact was located in this codebase at the time of writing, and we chose not to report a number we could not verify.

7. Limitations

This is a case study in how one principled domain-knowledge constraint failed on one task and one architecture; it is not a universal claim that domain-knowledge constraints cannot improve cross-hospital transfer. All experiments use a single clinical task (sepsis onset prediction) and a single transformer-based architecture and pretraining objective; whether the same confounds recur for other tasks, other architectures, or other domain-knowledge constraints is untested. Three random seeds is a small sample for a variance estimate, though the seed-variance finding itself (Section 5) is the point of that measurement, not a caveat to it.

References

Martin Arjovsky, Léon Bottou, Ishaan Gulrajani, and David Lopez-Paz. Invariant risk minimization. *arXiv preprint arXiv:1907.02893*, 2019.

Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171–4186, 2019.

Alistair E. W. Johnson et al. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, 10(1), 2023.

Tom J. Pollard et al. The eICU collaborative research database. *Scientific Data*, 5, 2018.

Matthew A. Reyna et al. Early prediction of sepsis from clinical data: The physionet/computing in cardiology challenge 2019. *Critical Care Medicine*, 48(2):210–217, 2020.

Chanu Rhee, Raymund Dantes, Lauren Epstein, David J. Murphy, Christopher W. Seymour, Theodore J. Iwashyna, Sameer S. Kadri, Derek C. Angus, Robert L. Danner, Anthony E. Fiore, et al. Incidence and trends of sepsis in us hospitals using clinical vs claims data, 2009–2014. *JAMA*, 318(13):1241–1249, 2017.

Shiori Sagawa, Pang Wei Koh, Tatsunori B. Hashimoto, and Percy Liang. Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization. In *International Conference on Learning Representations (ICLR)*, 2020.

Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio. Toward causal representation learning. *Proceedings of the IEEE*, 109(5):612–634, 2021.

Christopher W. Seymour, Vincent X. Liu, Theodore J. Iwashyna, Frank M. Brunkhorst, Thomas D. Rea, André Scherag, Gordon Rubenfeld, Jeremy M. Kahn, Manu Shankar-Hari, Mervyn Singer, et al. Assessment of clinical criteria for sepsis. *JAMA*, 315(8):762–774, 2016.

Mervyn Singer, Clifford S. Deutschman, Christopher Warren Seymour, Manu Shankar-Hari, Djillali Annane, Michael Bauer, Rinaldo Bellomo, Gordon R. Bernard, Jean-Daniel Chiche, Craig M. Coopersmith, et al. The third international consensus definitions for sepsis and septic shock (sepsis-3). *JAMA*, 315(8):801–810, 2016.

Nikhil Tamvada and Weizhi Lin. When validation lies: Diagnosing and fixing model selection failures under real hospital shift. submitted to IEEE URTC 2026, 2026.

Ashish Vaswani et al. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Andrew Wong, Erkin Otles, John P. Donnelly, Andrew Krumm, Jeffrey McCullough, Olivia DeTroyer-Cooley, Justin Pestrue, Marie Phillips, Judy Konye, Carleen Penoza, Muhammad Ghous, and Karandeep Singh. External validation of a widely implemented proprietary sepsis prediction model in hospitalized patients. *JAMA Internal Medicine*, 181(8):1065–1070, 2021. doi: 10.1001/jamainternmed.2021.2626.